

Q1.

List out foundation of AI and describe any two in detail.

Foundations of Artificial Intelligence

◆ **Definition**

Artificial Intelligence (AI) is an interdisciplinary field built on knowledge from various domains called **foundations of AI**.

◆ **Foundations of AI (List)**

1. Philosophy
 2. Mathematics
 3. Economics
 4. Neuroscience
 5. Psychology
 6. Computer Engineering
 7. Control Theory & Cybernetics
 8. Linguistics
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◆ **Explanation (Any Two)**

1. Mathematics

- Provides **formal methods and algorithms** for AI
- Helps in **problem solving and decision making**

Key Areas:

- Probability → handling uncertainty

- Linear Algebra → data representation
- Calculus → optimization

Applications:

- Machine Learning
- Neural Networks

Importance:

Ensures **accuracy and efficiency** in AI systems

2. Psychology

- Studies **human thinking and behavior**
- Helps AI systems **learn and act like humans**

Key Concepts:

- Learning
- Memory
- Problem-solving

Applications:

- Chatbots
- AI assistants
- Reinforcement learning

Importance:

Improves **human interaction and intelligence** in AI

◆ Short Notes (Other Foundations)

- **Philosophy:** Logic and reasoning
- **Economics:** Decision making, cost-benefit
- **Neuroscience:** Brain study → neural networks
- **Computer Engineering:** Hardware & software
- **Control Theory:** Automation and robotics
- **Linguistics:** Language processing (NLP)